Token-Efficient Context Sharing in Multi-Agent LLM Systems: How Much Can Be Cut, and What Limits It

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Abstract

Multi-agent LLM systems coordinate by forwarding context between agents, and every forwarded token is billed. Common practice forwards whole conversation histories, because deciding what to withhold is hard, so coordination cost grows quadratically in the number of agents. We ask a practical question: how much of that traffic can be removed without paying for it in task quality, and what stops us removing more?

On a public multi-hop benchmark reframed as a multi-receiver allocation problem, we find that a 2.7× reduction in inter-agent tokens costs no measurable accuracy, and that an oracle allocation reaches 21× while *improving* accuracy — so over-delivery is not merely wasteful but mildly harmful, and roughly an order of magnitude is available in principle. Pushing a practical policy to 5.4× does cost 8 accuracy points, so the frontier is real and we characterise it.

How the budget is spent matters more than how much is spent. Allocating per receiver beats selecting one globally-scored set for everyone by 19.2 accuracy points at matched cost. We explain this with a separation result: the one-set formulation — the natural reading of the problem as influence maximization — loses a factor factor n in the number of receivers, under two independent mechanisms, and does so even on instances where the objective is monotone and submodular; once saturation is strong enough the gap has no finite bound at all, so the factor n is that formulation's *best* case. We also show the two standard optimisation toolchains are unavailable here, the second because its structural parameters do not exist rather than because its bound is loose — and, on the positive side, that the correct per-receiver formulation costs nothing in approximability, since it decomposes exactly through a scalar budget price. Abandoning the one-set object is therefore not a tractability trade.

Finally, a negative result that we think is the most useful finding for practitioners. Across four allocation algorithms, including a resource-allocation program that exploits structure no baseline can, none beats a tuned uniform per-receiver rule at matched budget; and replacing lexical relevance with a dense retriever makes matters worse. Three independent lines of evidence place the binding constraint on *estimating* which agent needs what, not on optimising the allocation. The remaining order of magnitude is a retrieval problem, and effort spent on cleverer allocation will not recover it.

Keywords

multi-agent systems; large language models; communication; resource allocation; submodular optimization

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1 Introduction

A multi-agent LLM system coordinates by passing context. A planner tells a solver what the constraints are; an auditor is handed the records it must check. Every such transmission is billed per token, so coordination is a direct operating cost, and the prevailing engineering answer — forward the whole conversation, because deciding what to withhold is hard and sending everything is safe — makes that cost grow quadratically in the number of agents.

This paper treats that as the problem to solve. We ask how much inter-agent traffic can be removed before task quality suffers, what the shape of the cost–quality frontier is, and what prevents us from reaching the cheap allocations that clearly exist.

What we find. Three things, in decreasing order of how comfortable they are.

First, **a lot of the traffic is free to remove**. On a public multi-hop benchmark reframed as multi-receiver allocation, cutting inter-agent tokens by 2.7× costs no measurable accuracy, and an oracle allocation — which knows which item each agent needs — reaches 21× while *improving* accuracy by about five points. Over-delivery is therefore mildly harmful and not merely expensive, which is consistent with the measured tendency of LLMs to degrade under irrelevant context [18, 19, 25]. An order of magnitude is available in principle.

Second, **how the budget is spent matters more than how much**. Allocating a possibly different set to each receiver beats choosing one globally-scored set for everyone by 19.2 accuracy points at matched cost. This is not a tuning detail: we prove the one-set formulation loses a factor n in the number of receivers (Proposition 1), under two independent mechanisms, and even on instances where the objective is monotone and submodular — which turns out to be its best case, since the ratio becomes unbounded once saturation is strong enough to make broadcasting actively harmful. That formulation is the natural reading of context sharing as influence maximization [15], so a substantial and otherwise applicable literature [1, 7, 23] turns out to optimise the wrong object here. We also show that the nearest general framework for the messy objective this problem actually has is unavailable, because its structural parameters fail to exist rather than merely giving a loose bound (§5). The switch of object costs nothing in return: the per-receiver problem decomposes exactly through a scalar budget price, and any approximation available for one receiver transfers unchanged to n of them (Corollary 5).

Third, and most useful to anyone building such a system, **the remaining order of magnitude is not an optimisation problem**. We implement four allocation algorithms, including one that exploits separability structure no per-receiver baseline can imitate, and none beats a tuned uniform per-receiver rule at matched budget. Replacing lexical relevance with a dense retriever makes per-receiver delivery *worse*. Three independent observations converge: the binding constraint is estimating which agent needs what. The oracle fixes the scale — full delivery on 208 tokens where the best implementable policy reaches 62% on 929 — and no allocation algorithm closes that gap, because the information required is absent from the scores rather than badly exploited.

Contributions. (i) A measured cost–quality frontier for inter-agent context sharing, including the free reduction and the oracle ceiling (§7). (ii) A separation result showing the one-set formulation loses a factor n on its own best-case instances and is unbounded in general (§4), together with two degeneracy results removing the standard toolchains (§5). (iii) An exact decomposition of the per-receiver problem through a scalar budget price, with the consequence that its approximability equals the single-receiver one (§6). (iv) A negative result locating the residual gap in relevance estimation rather than allocation (§7). All theory statements were additionally checked by exhaustive enumeration on small instances.

2 Related work

Influence maximization. Monotonicity and submodularity of the spread function are what license greedy's $(1 - 1/e)$ [15], and the scaling literature [1, 7, 23] inherits those assumptions. Recent work relaxes them in various directions but keeps the seed-set object: one chosen set, serving the whole network. Our separation is orthogonal to the quality of those relaxations.

Non-monotone and non-submodular maximization. Guarantees exist for non-monotone submodular maximization [2, 10], for weakly submodular objectives via the submodularity ratio [5, 6] or supermodular degree [9], and for differences of submodular functions [13]. Closest to our setting, Shi and Lai [22] treat non-monotone, non-submodular maximization under a knapsack. §5 shows their parameters do not exist here. Regularized objectives of the form $g - c$ with g submodular and c modular admit distorted-greedy guarantees [12]; the density rule we discuss in §6 is an instance of that lineage, and we credit it as such rather than presenting it as new.

Context and communication cost in LLM agents. Empirical work cuts inter-agent tokens by pruning the message graph [26], generating the topology per task [8, 14], or sharing key–value caches instead of text [21]; none states an optimality guarantee, and a recent survey of token cost in agent systems reports no formal treatment of the allocation question either [3]. Closest in spirit to our separation, [20] studies empirically how correct and erroneous outputs propagate under topologies of differing sparsity, finding intermediate sparsity best — consistent with our claim that broadcasting is not the safe default, though reached without an optimization formulation. Closest to our applied setting, RCR-Router selects a memory subset per agent under a strict token budget and reports up to 30% savings on multi-hop QA, again without guarantees [17]. A parallel line does prove guarantees, but for a *single* receiver: PACMS treats context assembly as submodular selection [11], and BPS maximizes a monotone submodular benefit minus a modular token penalty under a hard budget, with a bicriteria $(1 - 1/e, 1)$ ratio [4]. BPS is the single-receiver, modular-penalty, submodular-benefit special case of the problem we study, and it is the closest prior work; it observes that redundant context "can even degrade performance" and then charges that degradation as a separate cost, keeping the benefit submodular. Both halves of the decomposition are where our setting departs. Its penalty $\kappa_E\,\ell(S)$ is linear in total length, with κ_E an explicitly *first-order* per-token sensitivity, whereas the degradation we measure accelerates with load (§3); and its benefit is a weighted coverage $\sum_k w_k h_k(\sum_{i\in S} u_{i,k})$ with each h_k concave, so items overlap within a dimension and are *additive* across dimensions. Two items that are separately inert but jointly decisive have no representation in that form, which is the case §5 turns on. On the theory side, [16] treats finite context windows as hard fan-in limits and locates saturation via a mixing depth, but through majority-vote aggregation rather than set-function optimization. We are not aware of a formal treatment of the multi-receiver token-allocation problem, nor of an influence-maximization framing of LLM-agent communication.

3 Model

Receivers (agents) are indexed $1, \ldots, n$. A pool F of information items is given, item f costing $c(f) > 0$ tokens. An *allocation* assigns a set $S_i \subseteq F$ to each receiver. Information is copyable, but each transmission is charged, so total spend is $\sum_i c(S_i)$ and serving n receivers with one set costs n times serving one. We also consider a *free-broadcast* convention, where a set may be given to everyone for one transmission, to show our results do not rest on the cost model.

Definition 1 (Multi-receiver context allocation). *An instance is a tuple $(n, F, c, (u_i)_{i\le n}, B)$ with n receivers, a finite item pool F, a cost $c : F \to \mathbb{R}_{>0}$ extended modularly to sets, per-receiver utilities $u_i : 2^F \to \mathbb{R}$ with $u_i(\emptyset) = 0$, and a budget $B > 0$. A feasible* allocation *is a tuple $(S_1, \ldots, S_n) \in (2^F)^n$ with $\sum_i c(S_i) \le B$, and the objective is $U(S_1, \ldots, S_n) = \sum_i u_i(S_i)$. The* seed-set restriction *of the same instance requires $S_1 = \cdots = S_n$.*

The seed-set restriction is exactly what an influence-maximization formulation imposes: one chosen set, serving the whole network. Definition 1 keeps everything else fixed, so the two formulations differ in the feasible region and nothing else — which is what makes the comparison in §4 meaningful rather than a change of subject.

Per-receiver utility is

$$u_i(S) = \mathrm{rel}_i(S) - \rho_i(\mathrm{conf}_i(S)), \tag{1}$$

relevance minus a saturation penalty. Two features of (1) are empirically motivated rather than assumed for convenience. First, ρ_i is increasing: delivering more context degrades an agent [19, 25]. Second, the penalty's argument is *confusable* load conf_i, not raw token count, because length-matched studies find that semantic proximity to the task predicts degradation while raw length barely does [18]. Cost and damage are therefore different quantities: one is billed and modular, the other is semantic and per-receiver. The objective is $U = \sum_i u_i(S_i)$ subject to the budget.

Example 1. Three agents share a pool of 20 documents. Agent 1 must check a delivery date, agent 2 a payment total, agent 3 must reconcile the two. Agents 1 and 2 each need one document; agent 3 needs both. A uniform allowance of one document per agent starves agent 3; an allowance of two overfeeds agents 1 and 2 with a document that can only distract them. No single set serves all three, and the cheapest correct allocation sends different sets to different agents. This is the structure Proposition 1 makes precise.

Why routing intuitions do not transfer. The problem looks like a delivery problem and is priced like one, which invites the vehicle-routing analogy. The analogy fails on the central property: goods are conserved, so a delivered unit is consumed and cannot serve a second customer, whereas context is *copyable* and the same item can serve every receiver at once. Conservation is what makes routing a partition problem; without it, the allocation is a covering problem in which the interesting constraint is not who gets the last unit but who should be spared it. Exactly one routing primitive does survive: *capacity*. A context window is rivalrous in the way a vehicle is — what one item occupies, another cannot — and that is the property (1) encodes through ρ_i. We therefore take capacity and discard conservation, and note that this is also why a per-vehicle-capacity formulation is not a route to the multi-receiver guarantee we lack: capacity here is soft, per-receiver, and semantic.

On the shape of ρ_i. We do not assume ρ_i convex, and this is deliberate: reported degradation curves are S-shaped, with a flat region followed by a steep fall, rather than convex from the origin [18, 19]. Magnitudes are also strongly task- and model-dependent, and at least one study finds large models almost unaffected by non-confusable filler. Any result of ours that needed a particular curvature would inherit that fragility. Proposition 4 is what buys the freedom: the penalty enters only through a chosen budget level, so ρ_i may be non-convex, discontinuous, or a cliff without affecting the decomposition. The one place curvature is needed is the density rule of §6, which reasons about ρ_i', and we flag it there as a limitation of that rule rather than of the model.

4 The seed-set formulation loses a factor n

Proposition 1. *For every $n \geq 1$ there is an instance with n receivers on which, at an identical budget,*

$$\frac{\mathrm{OPT}_{\textit{per-receiver}}}{\mathrm{OPT}_{\textit{seed-set}}} = n,$$

under either cost convention and for two independent reasons. Moreover the ratio is unbounded*: there are instances with $\mathrm{OPT}_{\textit{seed-set}} = 0 < \mathrm{OPT}_{\textit{per-receiver}}$.*

Proof. Take n receivers, n distinct items each of cost c, and let receiver i be satisfied only by item i. Set the budget $B = nc$.

(a) Billed per receiver. Per-receiver allocation sends item i to receiver i, spending $nc = B$ for utility n. A seed set S reaches all n receivers and so costs $n|S|c \leq nc$, forcing $|S| \leq 1$; a single item satisfies exactly one receiver, for utility 1.

(b) Free broadcast, saturating attention. Let the budget be non-binding, let a receiver absorb one item freely, and let each further item cost it 1. Broadcasting S to everyone yields $|S| - n\max(0, |S|-1)$, maximized at $|S| = 1$ with value 1, while per-receiver allocation again yields n.

(c) Unboundedness. Keep the same items and receivers, keep broadcast free, and let a receiver pay λ for each item it receives that is irrelevant to it. Broadcasting S gives $|S|\,(1 - \lambda(n-1))$, since every item satisfies its own receiver and distracts the other $n-1$. For any $\lambda > 1/(n-1)$ this is strictly negative for all $S \neq \emptyset$, so the best seed set is the empty one, of value 0. Per-receiver allocation delivers item i to receiver i alone, incurring no penalty at all, and again attains n. □

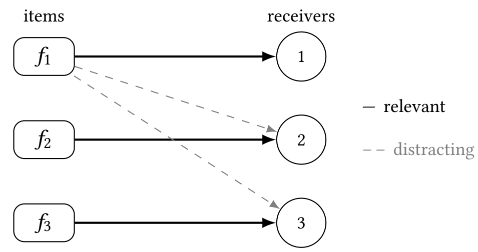

Figure 1: The separation instance of Proposition 1 at $n = 3$. Each item is relevant to exactly one receiver (solid) and distracting to the other $n-1$ (dashed, drawn only for f_1). A per-receiver allocation follows the solid edges and pays no penalty. A seed set must send the same items down the dashed edges too, so under (a) the budget confines it to one item and under (b)–(c) each added item helps one receiver and harms $n-1$. The number of dashed edges per item grows with n, which is the whole content of the result.

Case (b) is the more informative one. A reader may discount (a) as an artifact of charging per receiver; (b) removes that objection, because even with free broadcast the seed set is capped. The reason is worth stating plainly: *an additional broadcast item helps one receiver and harms the other $n-1$.*

Remark 1. The instance in case (a) contains no saturation and no complementarity: rel is modular and $\rho \equiv 0$, so the objective is monotone and submodular. The separation is therefore not a consequence of the degeneracies in §5. Repairing the objective's curvature does not rescue the seed-set formulation, because the loss comes from the choice of object, not from the objective's shape. Any guarantee proved against a seed-set optimum bounds distance from the wrong quantity.

Cases (a) and (c) bracket the result usefully. On the benign instance of (a) — modular relevance, no saturation, so the objective is monotone and submodular — the loss is exactly n and no more. Once saturation is strong enough to make broadcasting actively harmful, (c), the seed-set optimum is pinned at the empty allocation and the ratio admits no finite bound. The factor n is thus the *best case* for the one-set formulation, attained precisely where that formulation's own assumptions hold. Both constructions were checked by exhaustive enumeration over small instances rather than by algebra alone.

For completeness we record the hardness of the correct object; it is routine and we claim no credit for it.

Table 1: Existence of the two parameters that [22] requires *simultaneously*. Each phenomenon destroys exactly one.

	γ_1 (weak sub.)	γ_2 (weak super.)
Complementarity	**does not exist**	exists
Saturation	exists	**does not exist**

Proposition 2. *Maximising U subject to $\sum_i c(S_i) \le B$ is NP-hard, already for $n = 1$ and $\rho \equiv 0$.*

Proof. With one receiver and no saturation the problem is to choose $S \subseteq F$ with $c(S) \le B$ maximising a modular rel, which is Knapsack. For general n under one shared budget it contains Multiple Knapsack. ◻

Remark 2. What Proposition 2 does *not* settle is whether saturation adds hardness beyond the knapsack structure. The construction above switches ρ off, so the hardness is entirely attributable to the budget. We leave the question open and note that Proposition 5 below is evidence in the other direction: the multi-receiver structure adds no approximation hardness at all.

5 Both toolchains degenerate

Proposition 3. *On the class of instances of Definition 1 with* rel *modular and ρ increasing, U is neither monotone nor submodular; and no finite γ_1 and no $\gamma_2 \ge 1$ exist that make U simultaneously γ_1-weak submodular and γ_2-weak supermodular in the sense of [22].*

Monotonicity and submodularity fail. With rel modular and ρ increasing, an item whose relevance falls below its marginal saturation cost strictly lowers u_i, so U is not monotone. And two items individually inert but jointly decisive — $\mathrm{rel}(\emptyset) = \mathrm{rel}(\{a\}) = \mathrm{rel}(\{b\}) = 0$, $\mathrm{rel}(\{a, b\}) = 1$ — give increasing returns, so U is not submodular. Both are required by the greedy argument.

The nearest general framework has no parameters here. Shi and Lai [22] give guarantees for non-monotone, non-submodular maximization under a knapsack. Their general theorem requires the objective to be simultaneously γ_1-weak submodular and γ_2-weak supermodular:

$$F(U_2 \cup \{x\}) - F(U_2) \le \gamma_1 \left[F(U_1 \cup \{x\}) - F(U_1)\right], \quad (2)$$

$$\gamma_2 \left(F(U_2 \cup \{x\}) - F(U_2)\right) \ge F(U_1 \cup \{x\}) - F(U_1), \quad (3)$$

for $U_1 \subsetneq U_2$ and $x \notin U_2$. Our two phenomena destroy exactly one parameter each.

Under complementarity, take the instance above with $U_1 = \emptyset$, $U_2 = \{b\}$, $x = a$: (2) demands $1 \le \gamma_1 \cdot 0$, which no finite γ_1 satisfies. Under saturation, an item whose marginal is +0.5 at a lightly loaded receiver and −1.5 at a saturated one makes (3) demand $\gamma_2 \cdot (-1.5) \ge 0.5$, which no $\gamma_2 \ge 1$ satisfies.

Table 1 summarizes. The symmetry is the point: complementarity breaks weak *sub*modularity while saturation breaks weak *super*modularity, and since the theorem needs both at once it applies to neither phenomenon alone. This is a stronger statement than a loose bound — the parameters are undefined, so the guarantee is unavailable rather than weak. The witnesses above are the proof of Proposition 3, and were additionally found by exhaustive search over small instances, which is how we know the failure is not an artifact of a hand-picked example.

It is worth being clear about what this does *not* say. [22] is not thereby wrong or weak; its parameters are exactly the right ones for the class it addresses. The claim is only that our class sits outside that parameterisation, and it does so symmetrically: each of our two empirical phenomena annihilates one of the two parameters, so neither can be dropped and no single relaxation recovers the theorem. A framework that covered this setting would need to admit unbounded local curvature in both directions at once, which we do not know how to do.

6 Structure and algorithms

Proposition 1 says the per-receiver object is the right one. It does not say the resulting problem is intractable, and in one respect it is unusually well structured.

Proposition 4. *The problem is separable across receivers and coupled only by the shared budget. Consequently, writing $v_i(b) = \max\{u_i(S) : c(S) \le b\}$, an optimal allocation is obtained by solving*

$$\max \left\{ \textstyle\sum_i v_i(b_i) \ : \ \sum_i b_i \le B \right\}$$

and giving receiver i an optimal set at its level b_i. If each v_i is computed exactly, this is exact.

Proof. $U = \sum_i u_i(S_i)$ and the only constraint linking receivers is $\sum_i c(S_i) \le B$. Fix any feasible allocation and let $b_i = c(S_i)$. Then $u_i(S_i) \le v_i(b_i)$ and $\sum_i b_i \le B$, so its value is at most that of the displayed program. Conversely any feasible (b_i) with maximisers S_i is a feasible allocation of value $\sum_i v_i(b_i)$. ◻

Two consequences matter. First, the coupling between receivers is a single scalar: a budget price, recovered as the Lagrange multiplier of $\sum_i b_i \le B$. Second, and less obviously, the saturation penalty enters *only* through the chosen level b_i, so ρ_i may have any shape whatever — non-convex, discontinuous, a cliff. This matters because the measured degradation curves are S-shaped rather than convex, and a rule that reasons about ρ_i' requires convexity to argue that its admission bar rises as a receiver fills.

The decomposition also yields the one positive guarantee we have. It says that the difficulty of the multi-receiver problem is concentrated entirely in the single-receiver subproblems: whatever can be achieved for one receiver transfers, without loss, to n of them.

Corollary 5. *Suppose for each receiver i we have an oracle that, given a level b, returns a set $\widetilde{S}_i(b)$ with $c(\widetilde{S}_i(b)) \le b$ and $u_i(\widetilde{S}_i(b)) \ge (1 - \varepsilon)\, v_i(b)$. Then Algorithm 1, run on the curves $\widetilde{v}_i(b) = u_i(\widetilde{S}_i(b))$, returns a feasible allocation of value at least* $(1 - \varepsilon)$ OPT.

Proof. Let $(b_i^\star)$ maximise $\sum_i v_i(b_i)$ subject to $\sum_i b_i \le B$; by Proposition 4 its value is OPT. The algorithm returns $\max\{\sum_i \widetilde{v}_i(b_i) : \sum_i b_i \le B\}$, which is at least $\sum_i \widetilde{v}_i(b_i^\star) \ge (1 - \varepsilon) \sum_i v_i(b_i^\star) = (1 - \varepsilon)$ OPT. Feasibility and attainment hold because each $\widetilde{v}_i(b)$ is realised by an actual set of cost at most b, and $\widetilde{v}_i(b) \ge 0$ since $\emptyset$ is always available. ◻

Remark 3. The technique is standard resource allocation and we credit it as such; what is worth recording is the conclusion. With

Algorithm 1 Marginal allocation (exact given exact curves)

Require: receivers $1..n$, pool F, integer costs, budget B
for $i = 1$ to n **do**
  for $b = 0$ to B **do**
    $v_i(b) \leftarrow \max\{u_i(S) : c(S) \le b\}$ ▷ single-receiver oracle
  end for
end for
$A_0(b) \leftarrow 0$ for all b
for $i = 1$ to n **do**
  $A_i(b) \leftarrow \max_{0 \le t \le b} A_{i-1}(b-t) + v_i(t)$ ▷ for all $b \le B$
end for
recover the levels (b_i) attaining $A_n(B)$ **return** $(\widetilde{S}_1(b_1), \ldots, \widetilde{S}_n(b_n))$

$\rho_i \equiv 0$ each v_i is a knapsack value function and such oracles exist for every $\varepsilon > 0$, so in that regime the n-receiver problem is no harder to approximate than the 1-receiver one. Read alongside Proposition 1, the pair says something clean: the seed-set formulation is off by a factor n or worse, while the correct formulation costs nothing in approximability. The reason to abandon the one-set object is therefore not tractability.

Example 2. Take the three agents of the running example, each candidate document costing 1 token-unit, and let rel_i count the documents receiver i actually needs: v_1 and v_2 jump from 0 to 1 at $b = 1$ and stay flat, while v_3 stays at 0 until $b = 2$ and then jumps to 1, since agent 3 must reconcile both facts and one of them alone is worth nothing to it. At $B = 4$ the program splits $(1, 1, 2)$ for total value 3. Note what the flat regions do to the alternatives: a uniform allowance of $b_i = 4/3$ rounds to one document each and scores 2, because agent 3 gets a document it cannot use; raising the allowance to 2 each needs $B = 6$. A greedy density rule does no better, because at the moment it must choose, agent 3's first document has marginal value 0 and loses to agent 1's and agent 2's, which have marginal value 1 — and once the budget is spent the zero-marginal item is never revisited. The DP avoids this because v_3 is built for every level before any budget is committed, so the second document's value is visible at the time the split is chosen. This is the complementarity of §5 in its simplest form, and it is what the marginal program is designed to handle.

Two readings of the multiplier are worth recording. Operationally it is a price a system could publish: a scheduler that knows the current budget price can let each agent decide locally whether an item clears it, with no central allocation step, which matters in deployments where the orchestrator does not have the receivers' relevance estimates. Diagnostically, the multiplier's magnitude says how binding the budget is, and in our runs it is small over a wide middle range — which is the theoretical counterpart of the measured finding that a 2.7× reduction is free.

The second loop is $O(nB^2)$ arithmetic operations, or $O(nB)$ with the usual running-maximum bookkeeping when the v_i are concave; either way it is pseudo-polynomial in the budget and linear in the number of receivers, so the coupling across receivers is not where the cost lies. The first loop is the hard part, by Proposition 2. This is the sense in which the per-receiver object is well structured: the multi-receiver coupling is a scalar and the difficulty is local.

Two algorithm families follow, and we are explicit about what is new.

A density rule. The marginal of adding f to receiver i holding S is non-negative iff, to first order, $\Delta\mathrm{rel}_i(f \mid S)/c(f) \ge \rho_i'(\mathrm{conf}_i(S))$: send an item only when its relevance per token clears that receiver's current marginal degradation rate. An item failing the test both lowers u_i and consumes budget, so no optimal solution contains one — the restriction is a dominance property rather than an assumption. The threshold is a *dominance* property rather than a modelling assumption, and the argument is worth spelling out because it is the only place the model does any work for us. Suppose an optimal allocation gave receiver i an item f with $\Delta\mathrm{rel}_i(f \mid S_i \setminus \{f\}) < \rho_i' \cdot c(f)$ to first order. Deleting f raises u_i, because the relevance lost is smaller than the saturation charge avoided, and simultaneously frees $c(f) > 0$ of budget, which can only weaken the constraint. The resulting allocation is feasible and strictly better, contradicting optimality. So no optimum contains such an item, and the threshold restricts the search space without restricting the objective — unlike a cardinality cap, which does.

This rule is not new. Density-greedy with a positive-marginal filter is the distorted-greedy lineage [12], appears as Algorithm 1 of [22], and in the token setting has been applied to a single agent with a bicriteria guarantee [4]. The exchange argument above is likewise standard technique in that literature. We include the rule as a baseline for completeness and credit it accordingly; the two caveats we would attach are that the first-order reasoning needs ρ_i differentiable where the measured curves are S-shaped, and that a per-receiver threshold has no view of what a token would buy at another receiver — which is exactly what Algorithm 1 adds.

Marginal allocation. The objective is separable across receivers and coupled only by the shared budget, so the problem factors: build each receiver's value-versus-budget curve $v_i(b)$ and solve $\max \sum_i v_i(b_i)$ subject to $\sum_i b_i \le B$ by dynamic programming. This equalizes marginal value per token *across* receivers, which neither a uniform per-receiver rule nor a per-receiver density rule can do — the former spends identically on every receiver, the latter has no view of what a token would buy elsewhere. The saturation penalty enters only through the chosen level, so it may take any shape, including the non-convex ones the measurements suggest. §7 reports that this does not, in fact, outperform a tuned uniform rule on our benchmark, and why.

7 Experiments

Setup. We use MuSiQue-Ans [24], where the structure we need is annotated: each instance supplies 20 candidate paragraphs with support labels, and sub-questions each carrying the index of the paragraph answering it. Receivers are sub-questions, the item pool is the paragraph set, and cost is tokens transmitted summed over receivers. Ground-truth relevance being labeled makes an oracle allocation computable.

Two points of disclosure. MuSiQue is published as a single-agent task; splitting it across per-sub-question receivers is our framing. And most of its decompositions are sequential, so we restrict to

Table 2: Paired comparisons, $n = 120$ receiver-instances, matched-budget grid. Δ is the first policy minus the second; token ratio > 1 means the first is cheaper. Starred $p < 0.05$.

Comparison	Δacc	p	tokens
per-receiver vs. seed set	+0.192	0.0004*	1.1×
oracle vs. broadcast	+0.050	0.146	21.4×
allocation vs. broadcast	−0.042	0.227	2.7×
uniform k=5 vs. broadcast	−0.083	0.013*	5.4×
density vs. uniform (matched)	+0.000	1.000	0.8×

Table 3: Accuracy and cost per policy, matched-budget grid, $n = 120$. Tokens are per receiver; the ratio is broadcast's tokens divided by the policy's. Absolute exact-match is suppressed in favour of deltas because the benchmark is partially memorised (§7); 5% of receiver-instances are answered correctly with no context at all.

Policy	tokens	ratio	Δacc vs. broadcast
broadcast (all items)	2258	1.0×	—
oracle (labelled)	99	21.4×	+0.050
per-receiver, uniform	418	5.4×	−0.083
density rule	502	4.4×	−0.083
marginal allocation	831	2.7×	−0.042
seed set (matched)	260	8.7×	−0.263
random (matched)	345	6.5×	−0.632
no context	0	—	−0.763

the branching families in which two sub-questions carry no back-reference and are genuinely independent. Because the benchmark predates current models, some answers are memorized; an empty-allocation arm serves as a contamination control and we report policy *deltas* rather than absolute exact-match. Comparisons are paired — every policy answers the same items — using McNemar's exact test on discordant pairs and, for the token–accuracy points, a bootstrap over items.

Protocol. Two instruction-tuned models of the same family were used, at temperature 0, with a fixed answer-extraction template shared by every arm so that formatting cannot differ between policies. The matched-budget grid has 12 arms over 60 instances (1440 calls, all completed); the earlier 4-arm grid covers 69 instances at $n = 138$ receiver-instances. Every arm sees an identical prompt scaffold and differs only in which paragraphs the receiver is given, and arms are matched on delivered tokens before accuracy is compared, since an accuracy gain bought with a larger budget is not a gain from the mechanism. Runs are keyed and resumable per (instance, policy, arm, receiver), and duplicate keys are removed before analysis; without this the pairing silently breaks and n inflates.

The separation appears in the data. The strongest measured effect is the first row of Table 2: per-receiver allocation beats a seed-set baseline — items ranked by aggregate score and sent to everyone — by 19.2 accuracy points at matched cost. Proposition 1 predicts the

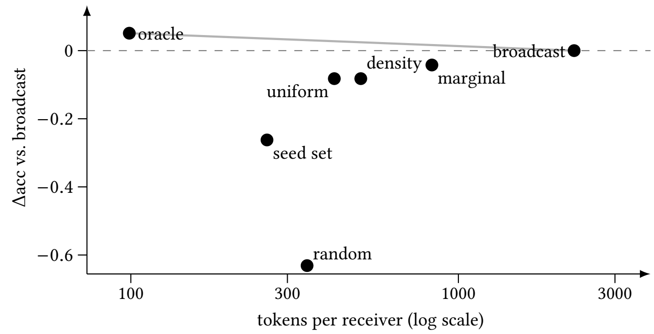

Figure 2: The measured cost–quality frontier, $n = 120$ receiver-instances. The dashed line is broadcast's accuracy; anything above it is both cheaper *and* better. The oracle sits there, at 21× fewer tokens, which is the order of magnitude the problem makes available. The implementable policies cluster well below the grey envelope, and — the point of the paper's negative result — they differ from one another by far less than any of them differs from the oracle. The seed set at matched cost is 26 points down, and random delivery (−0.632) fixes the scale of the task. An empty-allocation arm (not shown, 0 tokens, −0.763) is the contamination control.

Table 4: Proposition 1 predicts the seed-set gap grows with the number of receivers. Delivery recall as instances are fused to raise n. The seed set collapses; per-receiver allocation does not.

receivers n	2	8	16
seed set (matched cost)	0.45	0.08	0.01
per-receiver, uniform	0.67	0.56	0.56

gap should grow with the number of receivers, and it does: as we fuse instances to raise n from 2 to 8 to 16, seed-set delivery recall falls from 0.45 to 0.08 to 0.01 while per-receiver recall stays above 0.55.

Allocation is cheap. Allocating rather than broadcasting cuts tokens 2.7× with no measurable accuracy cost (row 3, $p = 0.227$). Pushing to 5.4× does cost 8 points, significantly (row 4), so the frontier is real. The oracle beats broadcast by $\sim$ 5 points at 21× fewer tokens, consistently across two models (+0.051 and +0.050) though significant on only one — we treat it as a consistent effect at borderline power rather than an established one.

A negative result on algorithm design. Across four value-curve designs, marginal allocation does not beat a tuned uniform per-receiver rule at matched budget; nor does the density rule (row 5, exactly zero at $p = 1.000$ while spending 25% more). We also found that replacing lexical relevance with a dense embedding scorer makes per-receiver retrieval *worse* (0.43 → 0.25 top-1 recall), because the sub-questions are terse relation templates whose entity identity a dense encoder discards — while the same change nearly doubles seed-set recall, which needs only instance-level topicality.

Table 5: Four designs for the value curve $v_i(b)$ that Algorithm 1 consumes, scored by delivery recall at matched spend against a tuned uniform per-receiver rule. The structure Proposition 4 exposes is real; on this benchmark it is not *estimable*.

Value curve $v_i(b)$	Outcome vs. tuned uniform
sum of relevance scores	**loses**: near-linear in b, so nothing to balance
noisy-OR over scores	ties; +4–5pp at large budgets only
captured own score mass	ties
any of the above, with composite receivers	ties ($\approx$ 11% cheaper at equal recall)

Table 6: Replacing lexical relevance with a dense embedding scorer *harms* per-receiver retrieval but helps the seed set. The two signals fail in opposite directions: embeddings capture what an instance is about, lexical matching captures which receiver needs what.

Delivery recall	lexical	embedding
per-receiver, top-1	**0.43**	0.25
per-receiver, top-5	**0.79**	0.61
seed set, k=3	0.45	**0.85**

Composite receivers were added specifically to manufacture the heterogeneity Proposition 1 draws its advantage from: the full multi-hop question needs *all* of its supporting paragraphs where a sub-question needs one, so a global k must starve one receiver or overfeed the other. Even then, marginal allocation only matches. The diagnosis is that a marginal rule can only help where the per-receiver curves are *distinguishable from label-free signals*, and here they are not: every receiver draws from a common pool with a similar score distribution, so an equal split is already near optimal. Heterogeneity in what receivers *need* did not surface as heterogeneity in what could be *estimated* about them.

The direction of that failure is legible in the data rather than mysterious. MuSiQue's branching sub-questions are mostly terse relation templates — of the form `"<entity> » <relation>"`, for instance `"The Poor Boob » screenwriter"` — and not natural-language questions. Exact entity-token overlap locates the right paragraph; a dense encoder maps such a string into a topical neighbourhood and discards the entity identity that distinguishes one receiver's need from another's. The seed-set row of Table 6 is the control that confirms this: it moves the other way, 0.45 $\longrightarrow$ 0.85, because sending one global set to everyone requires only instance-level topicality, which is exactly what a dense encoder captures well. So the two signals fail in opposite directions — embeddings know what an instance is about but not who needs what; lexical matching discriminates between receivers but misses paraphrase. The residual gap is therefore attributable to the *query representation* rather than to the scorer family, and a hybrid or entity-aware scorer is the obvious next attempt. We have not tried it, and we would not expect it to close the gap entirely.

Figure 2 puts all of this on one axis. The visual summary is the paper's argument in miniature: the gap between the implementable policies and the oracle dwarfs the gaps among the policies themselves, which is what it looks like when the binding constraint is not the allocation rule.

Three independent observations thus point the same way: the binding constraint is relevance *estimation*, not allocation *optimization*. The oracle fixes the scale of it, reaching full delivery on 208 tokens where the best implementable policy reaches 62% on 929. No allocation algorithm closes that gap, because the information required is not present in the scores.

8 Discussion

The results divide cleanly. The formulation-level claims are strong: the seed-set object is wrong by a factor $\Theta(n)$ even when the objective is well behaved, and both the classical toolchain and its nearest general successor are unavailable here. The algorithm-level claims are deliberately weak: we do not have a method that beats a tuned uniform per-receiver rule, and we report that rather than selecting a favorable budget point.

We take the negative result to be informative rather than merely disappointing. Marginal allocation can only help where per-receiver value curves are distinguishable from label-free signals, and on this benchmark they are not — every receiver draws from a common pool with a similar score distribution, so an equal split is already near optimal. Heterogeneity in what receivers *need* does not surface as heterogeneity in what can be *estimated* about them. Settings where it does — receivers with genuinely different and predictable appetites — are where the separation's practical value should be sought.

Threats to validity. We list these in the order we think a referee should weigh them.

The multi-receiver framing is ours. MuSiQue is published as a single-agent task, and splitting it into per-sub-question receivers is an interpretation, not a property of the dataset. It is a defensible one — the sub-questions and their supporting paragraphs are annotated, which is what makes the oracle computable — but no conclusion here should be read as a measurement on a deployed multi-agent system. We restrict to branching decompositions precisely because sequential ones would make the receivers dependent and the framing dishonest.

Contamination. The benchmark predates current models and some answers are memorised. Our control is an empty-allocation arm, on which 5% of receiver-instances are answered correctly with no context at all, and we therefore report policy deltas rather than absolute exact-match throughout. This does not remove contamination; it prevents it from being mistaken for policy quality, provided it affects arms equally, which pairing makes likely but does not guarantee.

Statistical power. The oracle-over-broadcast effect is +0.051 on one model and +0.050 on the other, and significant on only one (p = 0.039 versus p = 0.146). We report it as a consistent $\sim$ 5-point effect at borderline power and decline to call it established; a reader who saw only the significant model would be misled, and so would we.

Calibration of ρ. The saturation penalty's shape comes from published degradation curves whose magnitude is strongly model- and task-dependent, and at least one study finds large models essentially unaffected by non-confusable filler. Our theory needs degradation to exist in *some* regime rather than universally, and Proposition 1(a) does not need it at all; but the empirical claim about over-delivery is only as general as that literature.

Scope of the theory. Proposition 1 is a separation and Proposition 3 an impossibility. Neither supplies an approximation ratio for the multi-receiver problem with per-receiver saturation, and Corollary 5 supplies one only conditional on a single-receiver oracle. We regard the unconditional ratio as the main open question and have not solved it.

Open problems. Four, stated in what we judge to be decreasing order of tractability. (i) An unconditional approximation guarantee for Definition 1 with non-trivial ρ_i; Corollary 5 reduces this to the single-receiver case, where the obstruction is that u_i is neither monotone nor submodular, so the question is whether a receiver-local structural assumption weak enough to hold in practice suffices. (ii) Whether saturation adds hardness beyond the knapsack structure, which Proposition 2 leaves open by switching ρ off. (iii) Bundle identification: our complementarity witness assumes the jointly-decisive pair is known, whereas a system would have to discover it, and doing so is itself a search problem over the pool. (iv) Endogeneity of the graph. We treat the receiver set as given, but a deployed orchestrator decides which agents to instantiate in response to the task, so a transmission reshapes the topology it traverses. This is the assumption we are least comfortable with, and we see no route to a competitive-analysis treatment that does not first fix a model of how the orchestrator responds — which is an empirical question about a specific system, not a modelling choice we can make in the abstract.

9 Conclusion

Most inter-agent context traffic can be removed for free. A 2.7× reduction costs nothing measurable, and an oracle reaches 21× while improving accuracy, so the ceiling is roughly an order of magnitude rather than a few percent. That is the practical headline, and it holds on a public benchmark under paired testing.

Reaching the ceiling is a different matter, and the obstruction is not where we expected. Casting the problem as influence maximization is the natural move and the wrong one. The obstruction is not that agents degrade under irrelevant context, though they do, nor that information items interact, though they do; it is that the seed-set object — one set serving a network — is off by a factor n from the per-receiver object the problem actually poses, on instances where the objective is as well behaved as one could ask, and by an unbounded factor once it is not. Two toolchains that would otherwise apply are unavailable, the second because its structural parameters do not exist rather than because its bound is loose. And the switch of object is free: the per-receiver problem decomposes exactly, so it is no harder to approximate than its single-receiver special case.

But the route to the remaining order of magnitude does not run through better allocation algorithms — which is what we expected, and not what we found. It runs through better estimates of which receiver needs what. For a practitioner the implication is concrete: switch from broadcasting to per-receiver allocation, which is nearly free and worth 19 points over the globally-scored alternative, then spend the next increment of effort on retrieval rather than on the allocation rule.

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